Lines, angles, and triangles — Part 1

Geometry and Trigonometry · Math · 51 questions

1.1 4d53ed15-713b-4aa2-8943-17060404b617 · Easy

Triangles *ABC* and *DEF* are congruent, where *A* corresponds to *D*, and *B* and *E* are right angles. The measure of angle *A* is $18°$. What is the measure of angle *F*?

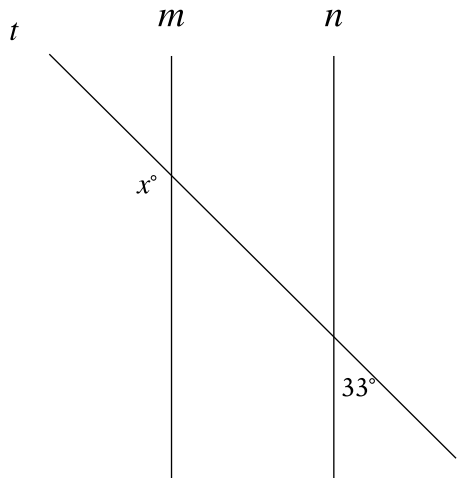
A. $18°$

B. $72°$

C. $90°$

D. $162°$

1.2 32a77034-9666-46e7-b78c-a2b1427d5d9b · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the lines are labeled t, m, and n.
- Line t intersects both line m and line n.
- At the intersection of line t and line m, 1 angle is labeled clockwise from top left as follows:
  - Bottom left: x°
- At the intersection of line t and line n, 1 angle is labeled clockwise from top left as follows:
  - Bottom right: 33°
- A note indicates the figure is not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

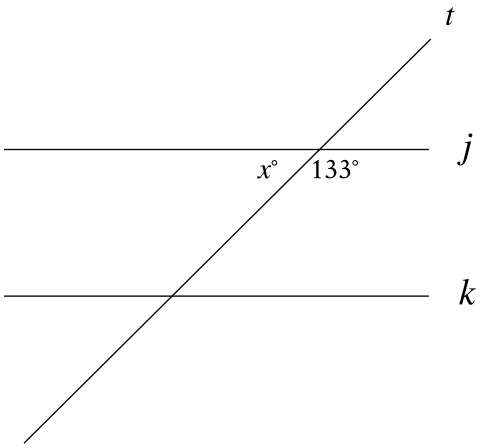
A. 33

B. 57

C. 123

D. 147

1.3 823ebf16-5273-4421-aeea-7ddb20cca4fd · Easy



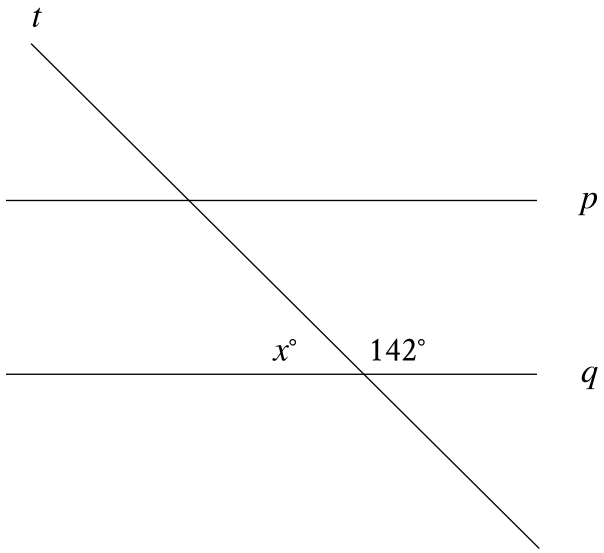
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, j, and k.
- Line t intersects both line j and line k.
- At the intersection of line t and line j, 2 angles are labeled clockwise from top left as follows:
  - Bottom right: 133°
  - Bottom left: x°
- A note indicates the figure is not drawn to scale.

In the figure, line *j* is parallel to line *k*. What is the value of *x*?

Student-produced response: _____

1.4 2e29358e-5d5c-485d-bea8-cb085eba5056 · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, p, and q.
- Line t intersects both line p and line q.
- At the intersection of line t and line q, 2 angles are labeled clockwise from top left as follows:
  - Top left: x°
  - Top right: 142°
- A note indicates the figure is not drawn to scale.

In the figure, line p is parallel to line q, and line t intersects both lines. What is the value of $x + 142$?

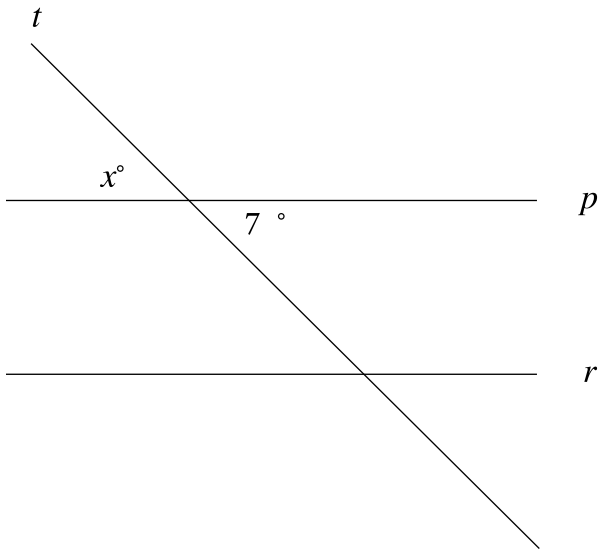
A. 52

B. 90

C. 142

D. 180

1.5 cbf53556-86ef-493b-a586-486445620deb · Easy



Note: . igure not drawn to scale.

- Clockwise from top left, the lines are labeled t, p, and r.
- Line t intersects both line p and line r.
- At the intersection of line t and line p, 2 angles are labeled clockwise from top left as follows:
  - Top left: x°
  - Bottom right: 72°
- Note: Figure not drawn to scale.

In the figure, line *p* is parallel to line *r*, and line *t* intersects both lines. What is the value of *x* ?

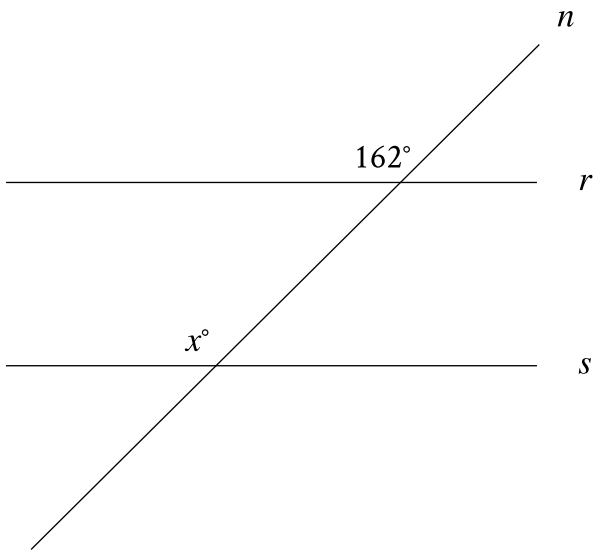
A. 36

B. 72

C. 180

D. 252

1.6 8d0aca2e-6014-42d2-adde-c7e271bac2be · Easy



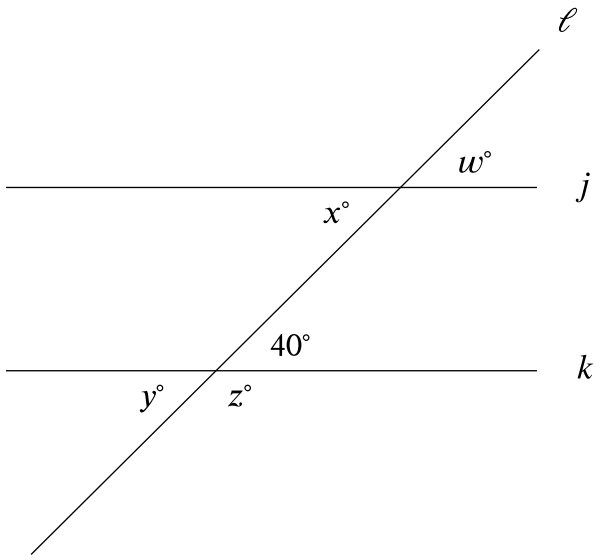
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled n, r, and s.
- Line n intersects both line r and line s.
- At the intersection of line n and line r, 1 angle is labeled clockwise from top left as follows:
  - Top left: 162°
- At the intersection of line n and line s, 1 angle is labeled clockwise from top left as follows:
  - Top left: x°
- A note indicates the figure is not drawn to scale.

In the figure, line *n* intersects lines *r* and *s*. Line *r* is parallel to line *s*. What is the value of *x* ?

Student-produced response: _____

1.7 af20e9d4-dd92-4ae0-9ead-3798a3e06b80 · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled script l, j, and k.
- Line script l intersects both line j and line k.
- At the intersection of line script l and line j, 2 angles are labeled clockwise from top left as follows:
  - Top right: w°
  - Bottom left: x°
- At the intersection of line script l and line k, 3 angles are labeled clockwise from top left as follows:
  - Top right: 40°
  - Bottom right: z°
  - Bottom left: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, line *l* intersects lines *j* and *k*. Which additional piece of information is sufficient to prove that lines *j* and *k* are parallel?

A. $w = 40$

B. $x = 140$

C. $y = 40$

D. $z = 140$

1.8 2b0e8838-4d1b-4eaf-9a92-439aa05ee874 · Easy

In a right triangle, the measure of one of the acute angles is $51°$. What is the measure, in degrees, of the other acute angle?

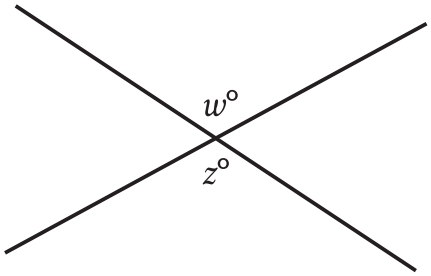
A. 6

B. 39

C. 49

D. 51

1.9 c6b681f6-504a-4e3a-8c73-a1cc8632d86b · Easy



Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from top as follows:
  - Top: w°
  - Right: unlabeled
  - Bottom: z°
  - Left: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z?

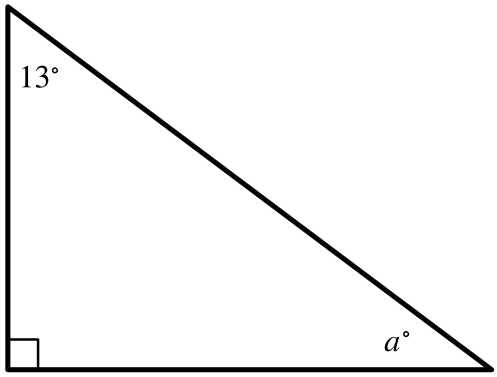
A. 36

B. 44

C. 68

D. 136

1.10 d14a2fdd-4d09-4911-b363-67cb2fe299e3 · Easy



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measures of the other 2 angles are as follows:
  - a°
  - 13°
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of *a*?

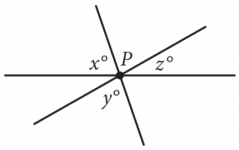
A. 13

B. 77

C. 90

D. 103

1.11 029285-DC · Easy



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

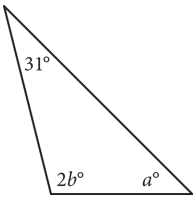
A. 140

B. 80

C. 40

D. 20

1.12 029330-DC · Easy



In the triangle above, $a = 45$. What is the value of b ?

A. 52

B. 59

C. 76

D. 104

1.13 b693eec7-7a41-4127-adf6-bb6bf263019b · Easy

In triangle ABC, $AB = 4{,}680$ millimeters mm and $BC = 4{,}680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

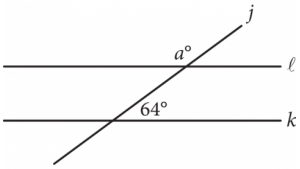
A. $AC = 4{,}680$ mm

B. $AC = 468$ mm

C. $AC = 46.8$ mm

D. $AC = 4.68$ mm

1.14 08250-DC · Easy



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

A. 26

B. 64

C. 116

D. 154

1.15 029526-DC · Easy

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

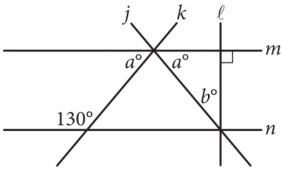
A. 10

B. 20

C. 30

D. 35

1.16 08194-DC · Easy



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

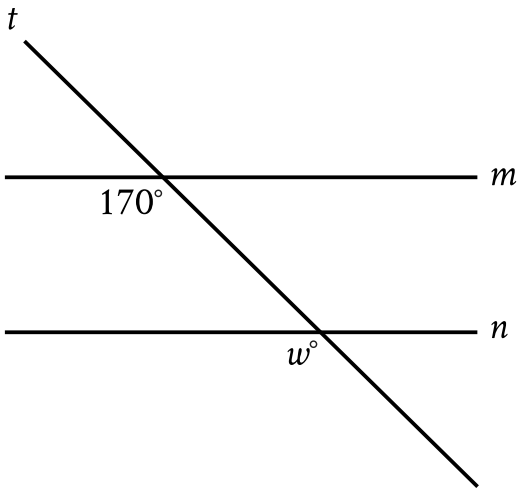
A. 40

B. 50

C. 65

D. 80

1.17 7bf2c7eb-d2eb-49db-a4cf-7cd376347a2c · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, m, and n.
- Line t intersects both line m and line n.
- At the intersection of line t and line m, 1 angle is labeled clockwise from top left as follows:
  - Bottom left: 170°
- At the intersection of line t and line n, 1 angle is labeled clockwise from top left as follows:
  - Bottom left: w°
- A note indicates the figure is not drawn to scale.

In the figure, line *m* is parallel to line *n*. What is the value of *w*?

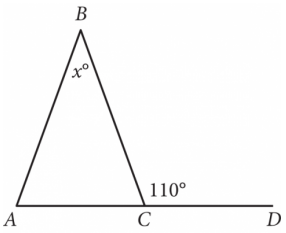
A. 17

B. 30

C. 70

D. 170

1.18 027075-DC · Easy



In the given figure, $\overline{AC}$ extends to point D. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

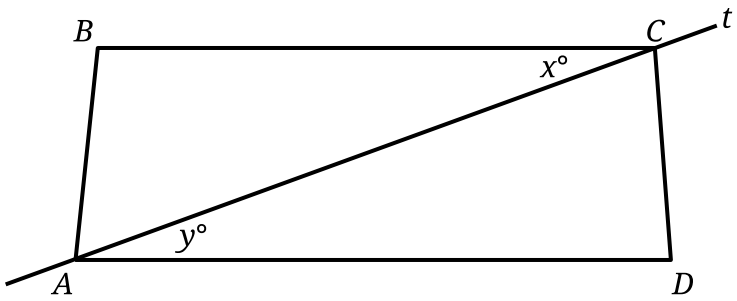
A. 110

B. 70

C. 55

D. 40

1.19 2d07d31c-7627-4c6e-813d-60c0e28346b7 · Easy



Note: Figure not drawn to scale.

- Line t passes through points upper A and upper C.
- The measure of angle upper A upper C upper B is x°.
- The measure of angle upper C upper A upper D is y°.
- A note indicates the figure is not drawn to scale.

In the figure shown, $\bar{BC}$ is parallel to $\bar{AD}$ and $x = 22$. Line t passes through points A and C. What is the value of y?

A. 22

B. 44

C. 68

D. 158

1.20 7d655176-161a-479c-b588-adb3cf6c69c2 · Easy

Triangles EFG and JKL are congruent, where E, F, and G correspond to J, K, and L, respectively. The measure of angle E is $45°$ and the measure of angle F is $20°$. What is the measure of angle J?

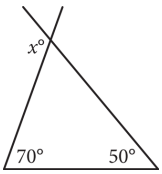
A. 20°

B. 45°

C. 135°

D. 160°

1.21 14890-DC · Easy



In the figure above, two sides of a triangle are extended. What is the value of x ?

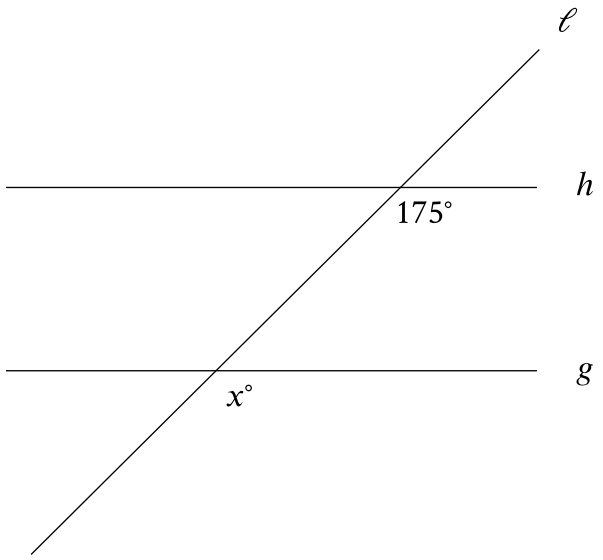
A. 110

B. 120

C. 130

D. 140

1.22 50ce06e1-35f6-477e-ac29-35d74d5ab3b6 · Easy



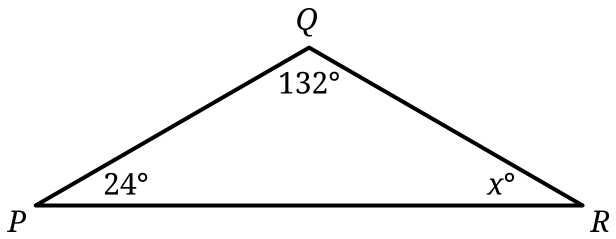
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled script l, h, and g.
- Line script l intersects both line h and line g.
- At the intersection of line script l and line h, 1 angle is labeled clockwise from top left as follows:
  - Bottom right: 175°
- At the intersection of line script l and line g, 1 angle is labeled clockwise from top left as follows:
  - Bottom right: x°
- A note indicates the figure is not drawn to scale.

In the figure shown, line g is parallel to line h. Line ℓ intersects both lines. What is the value of x?

Student-produced response: _____

1.23 96a4ec26-6ad9-4b3b-9ae1-693feee5691e · Easy



Note: Figure not drawn to scale.

- The measure of angle upper P is 24°.
- The measure of angle upper Q is 132°.
- The measure of angle upper R is x°.
- A note indicates the figure is not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x?

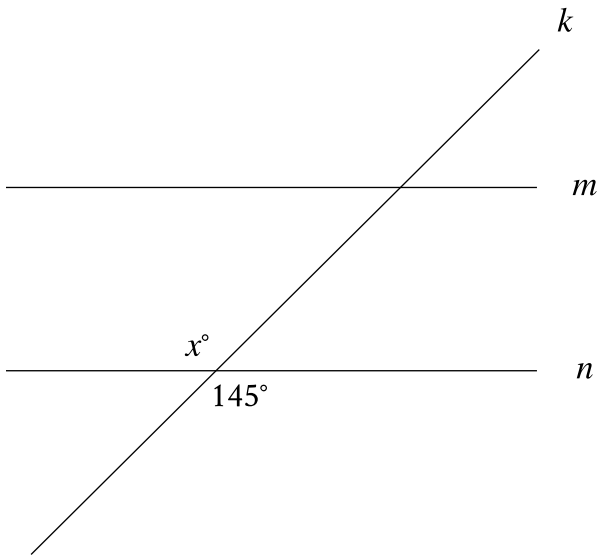
A. 156

B. 66

C. 48

D. 24

1.24 af58075c-8456-4696-8179-6c1a86044478 · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled k, m, and n.
- Line k intersects both line m and line n.
- At the intersection of line k and line n, 2 angles are labeled clockwise from top left as follows:
  - Top left: x°
  - Bottom right: 145°
- A note indicates the figure is not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *k* intersects both lines. Which of the following statements is true?

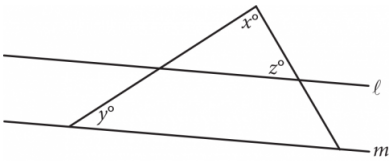
A. The value of *x* is less than 145.

B. The value of *x* is greater than 145.

C. The value of *x* is equal to 145.

D. The value of *x* cannot be determined.

1.25 019566-DC · Easy



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

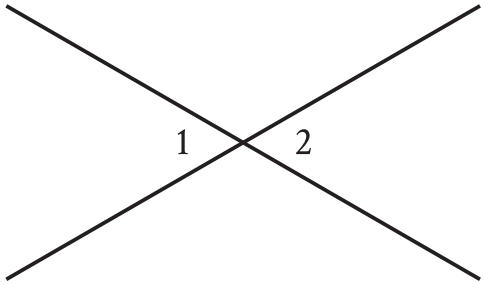
A. 120

B. 100

C. 90

D. 80

1.26 a79dbc7d-8d01-40c4-8900-15940eeff7d7 · Easy



Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from left as follows:
  - Left: 1
  - Top: unlabeled
  - Right: 2
  - Bottom: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. Angle 1 and angle 2 are vertical angles. The measure of angle 1 is $72°$. What is the measure of angle 2?

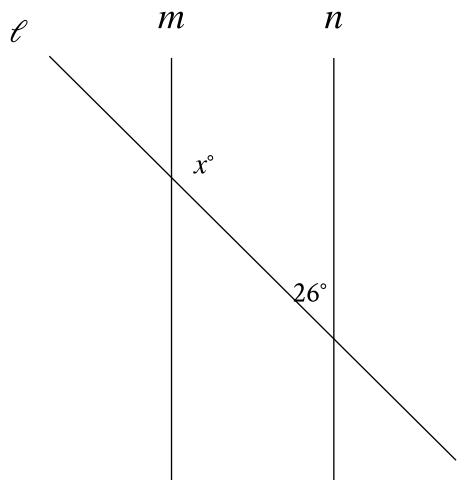
A. $72°$

B. $108°$

C. $144°$

D. $288°$

1.27 095e8ea4-b798-4679-8448-1de5fc94090d · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled l, m, and n.
- Line l intersects both line m and line n.
- At the intersection of line l and line m, 1 angle is labeled clockwise from top left as follows:
  - Top right: x°
- At the intersection of line l and line n, 1 angle is labeled clockwise from top left as follows:
  - Top left: 26°
- A note indicates the figure is not drawn to scale.

In the figure shown, line *m* is parallel to line *n*. What is the value of *x*?

A. 13

B. 26

C. 52

D. 154

1.28 5ca08c31-e18f-4cb7-837a-0814ebea3a8e · Easy

In $\triangle RST$, the measure of $\angle R$ is $63°$. Which of the following could be the measure, in degrees, of $\angle S$?

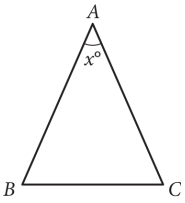
A. 116

B. 118

C. 126

D. 180

1.29 026874-DC · Easy



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of $67°$. What is the value of x ?

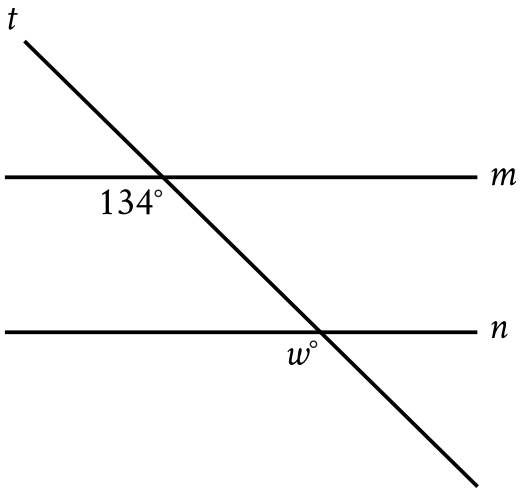
A. 36

B. 46

C. 58

D. 70

1.30 9df9e1c7-bfff-4d4e-865f-5cbcb4b24e4c · Easy



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, m, and n.
- Line t intersects both line m and line n.
- At the intersection of line t and line m, 1 angle is labeled clockwise from top left as follows:
  - Bottom left: 134°
- At the intersection of line t and line n, 1 angle is labeled clockwise from top left as follows:
  - Bottom left: w°
- A note indicates the figure is not drawn to scale.

In the figure, line *m* is parallel to line *n*. What is the value of *w*?

A. 13

B. 34

C. 66

D. 134

1.31 baefd1d0-91cf-4b82-9c41-2145510636af · Easy

In $\triangle XYZ$, the measure of $\angle X$ is $24°$ and the measure of $\angle Y$ is $98°$. What is the measure of $\angle Z$?

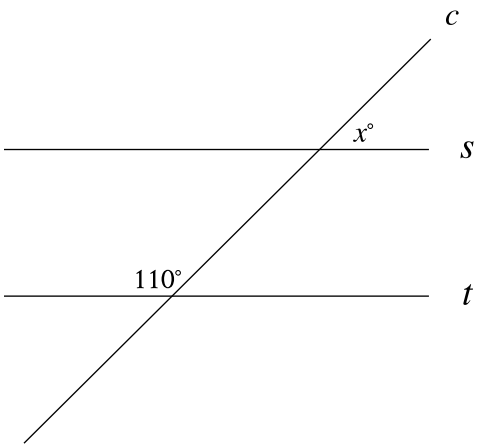
A. $58°$

B. $74°$

C. $122°$

D. $212°$

1.32 b09e0918-6dfc-4b50-b503-bbdd89fe3480 · Easy



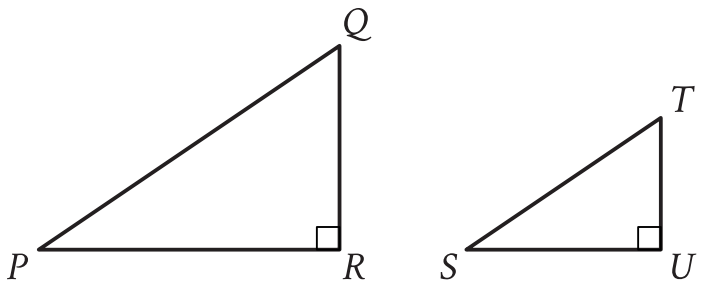
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled, c, s, and t.
- Line c intersects both line s and line t.
- At the intersection of line c and line s, 1 angle is labeled clockwise from top left as follows:
  - Top right: x°
- At the intersection of line c and line t, 1 angle is labeled clockwise from top left as follows:
  - Top left: 110°
- A note indicates the figure is not drawn to scale.

In the figure shown, line *c* intersects parallel lines *s* and *t*. What is the value of *x*?

Student-produced response: _____

1.33 dbae1ccb-c727-453d-aad3-d9808097dcfd · Easy



Note: Figures not drawn to scale.

- Right triangle upper P upper Q upper R is labeled as follows:
  - Angle upper R is a right angle.
- Right triangle upper S upper T upper U is labeled as follows:
  - Angle upper U is a right angle.
- A note indicates the figures are not drawn to scale.

Right triangles *PQR* and *STU* are similar, where *P* corresponds to *S*. If the measure of angle *Q* is $18°$, what is the measure of angle *S*?

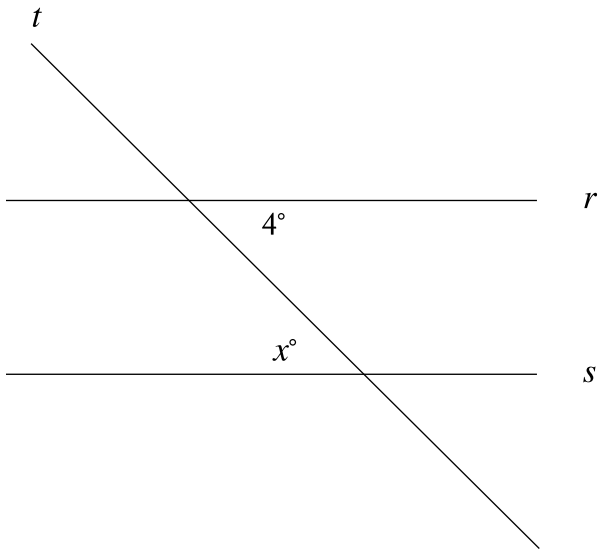
A. $18°$

B. $72°$

C. $82°$

D. $162°$

1.34 3f1bac86-dd93-4b6b-960c-c29978b05446 · Easy



Note: . igure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, r, and s.
- Line t intersects both line r and line s.
- At the intersection of line t and line r, 1 angle is labeled clockwise from top left as follows:
  - Bottom right: 24°
- At the intersection of line t and line s, 1 angle is labeled clockwise from top left as follows:
  - Top left: x°
- Note: Figure not drawn to scale.

In the figure, line *t* intersects lines *r* and *s*. Which additional piece of information is sufficient to prove that lines *r* and *s* are parallel?

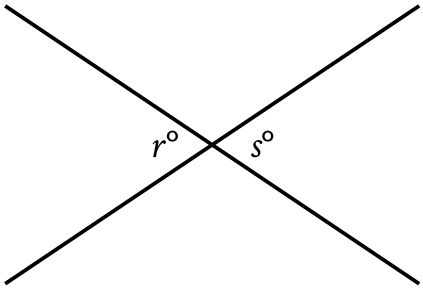
A. $x < 24$

B. $x = 24$

C. $x > 24$

D. $x - 24 = 360$

1.35 bafc5567-f967-4844-ba8a-9cd3d5bc11de · Easy



Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from left as follows:
  - Left: r°
  - Top: unlabeled
  - Right: s°
  - Bottom: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. If $r = 53$, what is the value of s?

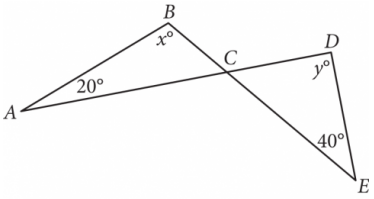
A. 37

B. 53

C. 143

D. 233

1.36 022230-DC · Easy



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

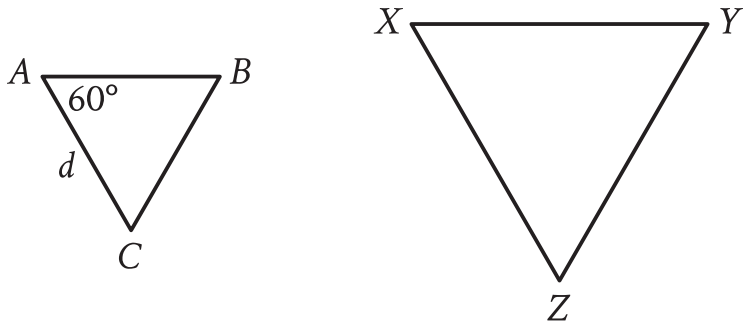
A. 100

B. 90

C. 80

D. 60

1.37 c6d50efa-46fb-4406-9c84-8c8d2ab9e927 · Easy



Note: Figures not drawn to scale.

- The 2 triangles are oriented such that upper A corresponds to upper X, upper B corresponds to upper Y, and upper C corresponds to upper Z.
- Triangle upper A upper B upper C is labeled as follows:
  - The measure of angle upper A is 60°.
  - The length of side upper A upper C is d.
- A note indicates the figures are not drawn to scale.

For the triangles shown, triangle *ABC* is dilated by a scale factor of 3 to obtain triangle *XYZ*, where $d = 16$. What is the measure, in degrees, of angle *X*?

A. 20

B. 57

C. 60

D. 63

1.38 60ddced0-ee94-414c-9143-3704d3c18d2e · Easy

In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal and the measure of $\angle L$ is $118°$. What is the measure of $\angle J$?

A. 31°

B. 59°

C. 62°

D. 90°

1.39 b3738b72-9c5a-45b5-986b-00bb9643ab90 · Easy

In $\triangle XYZ$, the measure of $\angle X$ is $23°$ and the measure of $\angle Y$ is $66°$. What is the measure of $\angle Z$?

A. $43°$

B. $89°$

C. $91°$

D. $179°$

1.40 6145108f-5273-4b08-afde-8e75072e2eef · Easy

In triangle ABC, the measure of angle B is $52°$ and the measure of angle C is $17°$. What is the measure of angle A?

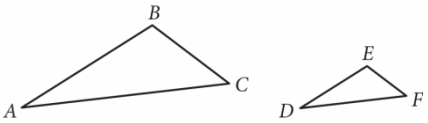
A. $21°$

B. $35°$

C. $69°$

D. $111°$

1.41 027209-DC · Medium

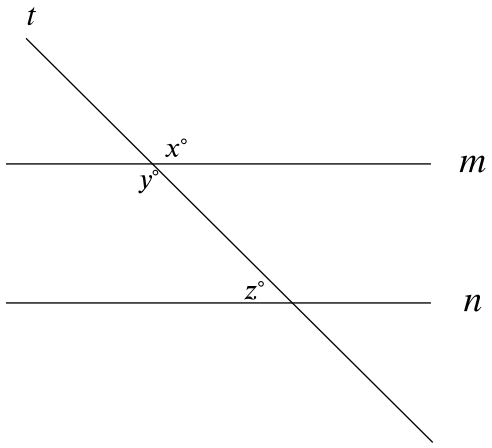


Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20°, what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

Student-produced response: _____

1.42 551e10b5-2f0d-4002-a399-823a818766b9 · Medium



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, m, and n.
- Line t intersects both line m and line n.
- At the intersection of line t and line m, 2 angles are labeled clockwise from top left as follows:
  - Top right: x°
  - Bottom left: y°
- At the intersection of line t and line n, 1 angle is labeled clockwise from top left as follows:
  - Top left: z°
- A note indicates the figure is not drawn to scale.

In the figure, lines *m* and *n* are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z?

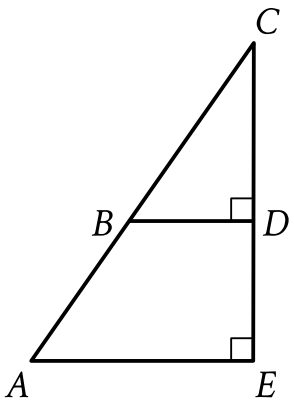
A. 3

B. 21

C. 41

D. 139

1.43 6a820c4b-cc06-4343-a5bd-fe398a0d2ec6 · Medium



Note: Figure not drawn to scale.

- Line segment upper B upper D is drawn from side upper C upper A to side upper C upper E to form triangle upper C upper B upper D.
- Angle upper C upper D upper B is a right angle.
- Angle upper C upper E upper A is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, triangle *CAE* is similar to triangle *CBD*. The measure of angle *CBD* is $57°$, and $AE = 26BD$. What is the measure of angle *CAE*?

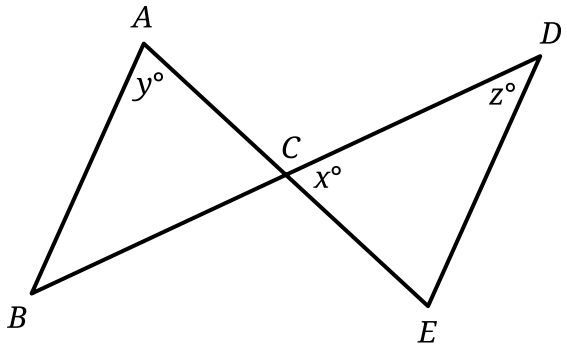
A. $26 \cdot 57°$

B. $26 + 57°$

C. $57°$

D. $26°$

1.44 b1f0e2ca-1e0c-430d-b61d-6cf8594501a5 · Medium



Note: Figure not drawn to scale.

- Triangle upper A upper B upper C has a common vertex with triangle upper C upper D upper E.
- The common vertex is upper C.
- Triangle upper A upper B upper C is labeled as follows:
  - The measure of angle upper B upper A upper C is y°.
- Triangle upper C upper D upper E is labeled as follows:
  - The measure of angle upper D upper C upper E is x°.
  - The measure of angle upper C upper D upper E is z°.
- A note indicates the figure is not drawn to scale.

In the figure, $\bar{AE}$ intersects $\bar{DB}$ at point C and $\bar{AB}$ is parallel to $\bar{DE}$, creating triangles ABC and CDE. If $x = 63$ and $y = 2x - 47$, what is the value of z?

A. 38

B. 47

C. 63

D. 79

1.45 64da6245-4314-4da9-817e-46dcdd9dc29d · Medium

Right triangles *LMN* and *PQR* are similar, where *L* and *M* correspond to *P* and *Q*, respectively. Angle *M* has a measure of $53°$. What is the measure of angle *Q*?

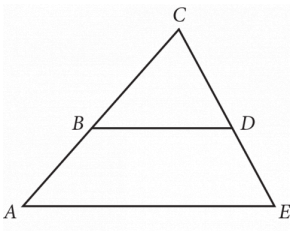
A. $37°$

B. $53°$

C. $127°$

D. $143°$

1.46 029403-DC · Medium



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62°, what is the measure of angle CAE ?

A. 58°

B. 60°

C. 62°

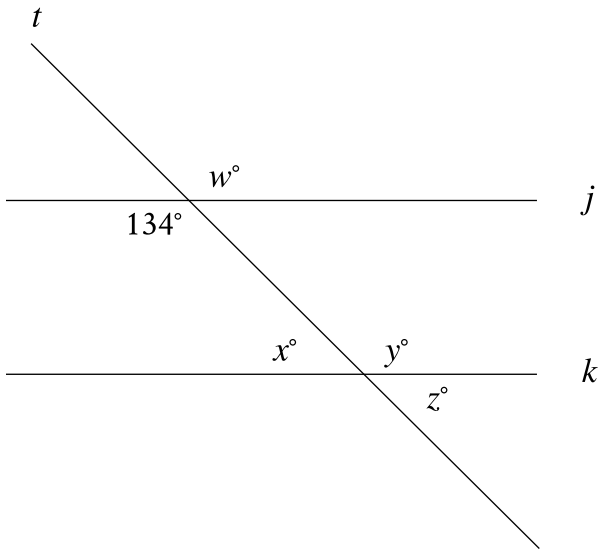
D. 120°

1.47 e8922caa-569e-45e5-9744-1704932fd187 · Medium

Triangles *ABC* and *DEF* are congruent, where *A* corresponds to *D*, and *B* and *E* are right angles. The measure of angle *A* is $69°$. What is the measure, in degrees, of angle *F*?

Student-produced response: _____

1.48 275a2867-6329-41e4-8bab-072a1e2475e7 · Medium



Note: . igure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, j, and k.
- Line t intersects both line j and line k.
- At the intersection of line t and line j, 2 angles are labeled clockwise from top left as follows:
  - Top right: w°
  - Bottom left: 134°
- At the intersection of line t and line k, 3 angles are labeled clockwise from top left as follows:
  - Top left: x°
  - Top right: y°
  - Bottom right: z°
- Note: Figure not drawn to scale.

In the figure, line *t* intersects lines *j* and *k*. Which additional piece of information is sufficient to prove that lines *j* and *k* are parallel?

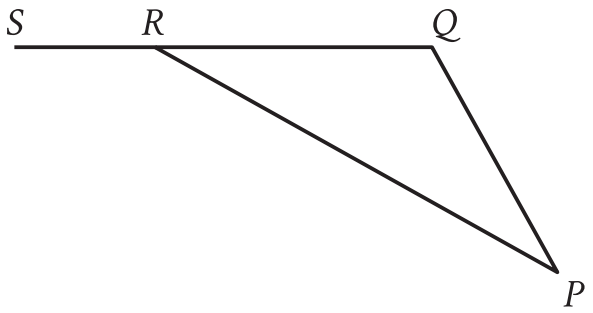
A. $x = 46$

B. $y = 46$

C. $w = 134$

D. $z = 134$

1.49 2fc71dbf-343a-44f3-80ee-5b7f22a564da · Medium



Note: Figure not drawn to scale.

- Line segment upper Q upper R is extended to point upper S.
- A note indicates the figure is not drawn to scale.

In triangle PQR, QR is extended to point S. The measure of $\angle PQR$ is $132°$, and the measure of $\angle PRS$ is $163°$. What is the measure of $\angle QPR$?

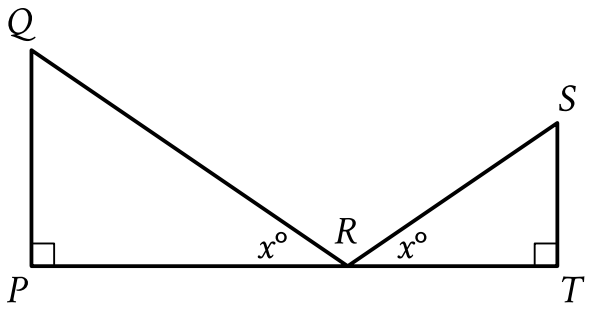
A. $48°$

B. $31°$

C. $24°$

D. $17°$

1.50 9fe57ebf-964f-4add-9e16-f54cf44c7bf4 · Medium



Note: Figure not drawn to scale.

- The 2 triangles share common vertex upper R.
  - Common vertex upper R lies on line segment upper P upper T.
- Triangle upper Q upper P upper R is labeled as follows:
  - Angle upper Q upper P upper R is a right angle.
  - The measure of angle upper Q upper R upper P is x°.
- Triangle upper S upper T upper R is labeled as follows:
  - Angle upper S upper T upper R is a right angle.
  - The measure of angle upper S upper R upper T is x°.
- A note indicates the figure is not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\bar{ST}$, $\bar{QP}$, $\bar{PR}$, and $\bar{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\bar{SR}$?

A. $\frac{350}{15}$

B. $\frac{350}{20}$

C. $\frac{210}{20}$

D. $\frac{210}{25}$

1.51 e2fa4379-f216-4286-8c00-6098b985ceea · Medium

Triangle ABC is similar to triangle XYZ, where A, B, and C correspond to X, Y, and Z, respectively. In triangle ABC, the length of $\bar{AB}$ is 170 and the length of $\bar{BC}$ is 850. In triangle XYZ, the length of $\bar{YZ}$ is 60. What is the length of $\bar{XY}$?

A. 204

B. 182

C. 60

D. 12